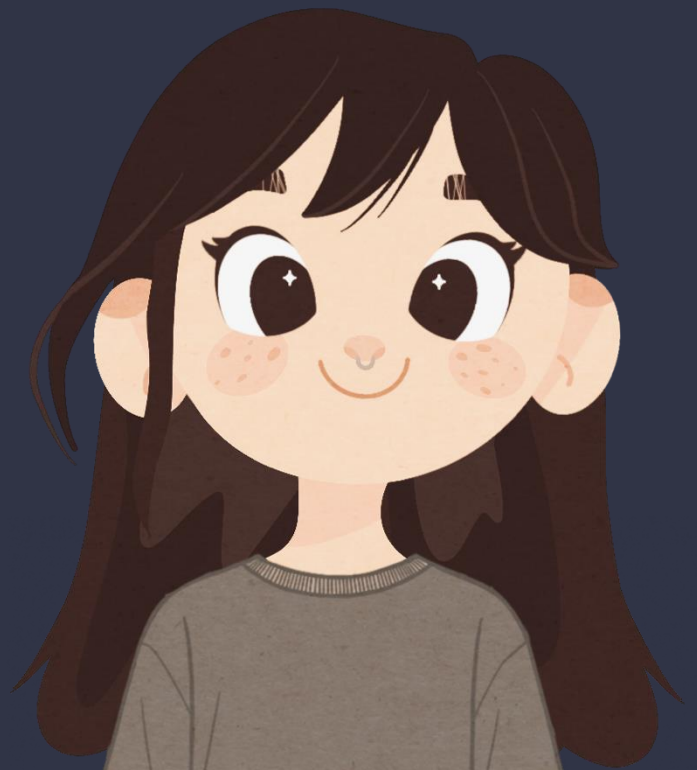
Becoming a code review champ



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Lead engineer @ Disney

As a code author…

Before committing

Use VSCode’s Source Control to view every diff, file by file

> *What can I clean up or rewrite?*

When opening a new PR



Write PR description and open the PR ***as a draft***



Do a self-review - “What can I clean up or rewrite?”



Push new commit with any fixes before asking for PR review



Leave behind comments to help guide reviewers (helpful for large or complex PRs)

PR descriptions

Goal

Make it as easy as possible for someone without context on the problem you're solving to review your code



What is being changed and *why*



Before & after screenshots



Resource links



Testing gotchas

When opening a new PR



Write PR description and open the PR ***as a draft***



Do a self-review - “What can I clean up or rewrite?”



Push new commit with any fixes before asking for PR review



Leave behind comments to help guide reviewers (helpful for large or complex PRs)

Helpful self-review comments



Areas reviewers should pay close attention to



Why a certain change was made



Reminders for yourself for later



Questions for reviewers

As a code reviewer…

As a code reviewer…

Your job is to look at every line of code

If you don't feel qualified to review some parts, that's okay!

Make sure there's a reviewer on the PR who is qualified, particularly for complex issues like privacy, security, concurrency, accessibility, internationalization, …

As a code reviewer…

You’re not just looking for code improvements.

Calling out good practices, problem-solving, etc is valuable for mentoring 

As a code reviewer…

Favor approving a PR when it's in a state where it improves overall code health, even if it isn't perfect.

There's no such thing as "perfect" code – only *better* code. Trade offs and different approaches will always exist.

Perfect is the enemy of done. Balance forward progress with the importance of suggested changes. Create follow up tasks as needed *as long as it doesn't compromise code quality.*

What to look for during code review

- Code is well-designed – architecture, *where* the code lives, follows existing patterns
- Functionality is correct – edge cases, scalability, etc
- UI changes match specs, follow a11y, etc
- Code isn't more complex than necessary
- Appropriate and well-designed tests that meet coverage requirements
- Comments are clear and useful, explaining *why* instead of *what*

More reading
Google's engineering practices

How do you actually do this?

Pass 1

Overall look to understand approach, architecture, design

Good time to leave comments if something is unclear.

If there’s major issues here, they should be surfaced quickly so the dev can pivot.

How do you actually do this?

Pass 2

Close look line-by-line for code quality, edge cases, style conformation, alignment with specs, …

How do you actually do this?

Pass 3

Close look at tests, documentation, non-user-facing pieces

Guiding principles

Be respectful



“Why would you use threads here when there’s obviously no benefit gained from concurrency?”



“This is adding complexity without any performance benefit I can see. If there’s no perf benefit, let’s just use a single-threaded approach.”

Guiding principles

Provide actionable feedback and explain *why*



"This isn't an optimal approach."



"This could have scalability issues. Let's refactor so all promises are run in parallel, instead of waiting for one to finish before calling the next."

Guiding principles

Balance giving guidance vs pointing out issues



Primary goal is to make sure code quality is high.



Secondary goal is learning and improving dev skills.

Guiding principles

Label comment severity



“nit: remove comment”

“question: do we also need to account for X here?”

“consider: it might be worth adding telemetry here to capture the error”

Guiding principles

Dedicate heads-down time for code review



Do what works for you but aim to have dedicated time every day to review code without context switching.



Block off time on your calendar

thank you

